x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1891	98,752				
		33,533			
1901	1,32,285		2,258		
		35,791		–10,435	
1911	1,68,076		-8,177		41,358
		27,614		30,293	
1921	1,95,690		22,746		
		50,360			
1931	2,46,050				

$$y_{(x=1905)} = 98,752 + (1.4)(33533) + \frac{(1.4)(0.4)}{2}(2258) + \frac{(1.4)(0.4)(-0.6)}{6}(-10435) + \frac{(1.4)(0.6)(-0.6)(-1.6)}{24}(41358)$$

= 98752 + 46946.2 + 632.24 + 584.36 + 1389.63

= 148304.43

≅ 1,48,304

Example 5.14

The values of $y = f(x)$ for $x = 0,1,2, \ldots,6$ are given by

x	0	1	2	3	4	5	6
y	2	4	10	16	20	24	38

Estimate the value of y (3.2) using forward interpolation formula by choosing the four values that will give the best approximation.

Solution:

Since we apply the forward interpolation formula,last four values of $f(x)$ are taken into consideration (Take the values from x = 3).

The forward interpolation formula is

$$y_{(x=x_0+nh)} = y_0 + \frac{n}{1!}\Delta y_0 + \frac{n(n-1)}{2!}\Delta^2 y_0 + \frac{n(n-1)(n-2)}{3!}\Delta^3 y_0 + \ldots$$

$x_0 + nh = 3.2, x_0 = 3, h = 1$

$\therefore \quad n = \frac{1}{5}$

The difference table is

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
3	16			
		4		
4	20		0	
		4		10
5	24		10	
		14		
6	38			

$$y_{(x=3.2)} = 16 + \frac{1}{5}(4) + \frac{\frac{1}{5}\left(\frac{-4}{5}\right)}{2}(0) + \frac{\frac{1}{5}\left(\frac{-4}{5}\right)\left(\frac{-9}{5}\right)}{6} \times 10$$

$= 16 + 0.8 + 0 + 0.48 = 17.28$

Example 5.15

From the following table find the number of students who obtained marks less than 45.

Marks	30-40	40-50	50-60	60-70	70-80
No. of Students	31	42	51	35	31

Solution:

Let x be the marks and y be the number of students.

By converting the given series into cumulative frequency distribution, the difference table is as follows.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
Less than 40	31				
		42			
50	73		9		
		51		–25	
60	124		-16		37
		35		12	
70	159		-4		
		31			
80	190				

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